

Python Project

Hotel Management System

Group Members:

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- A Desktop-Based Application
- Developed using Python, Tkinter & SQLite
- Using different classes and functions
- With basic understanding of OOPS in python.

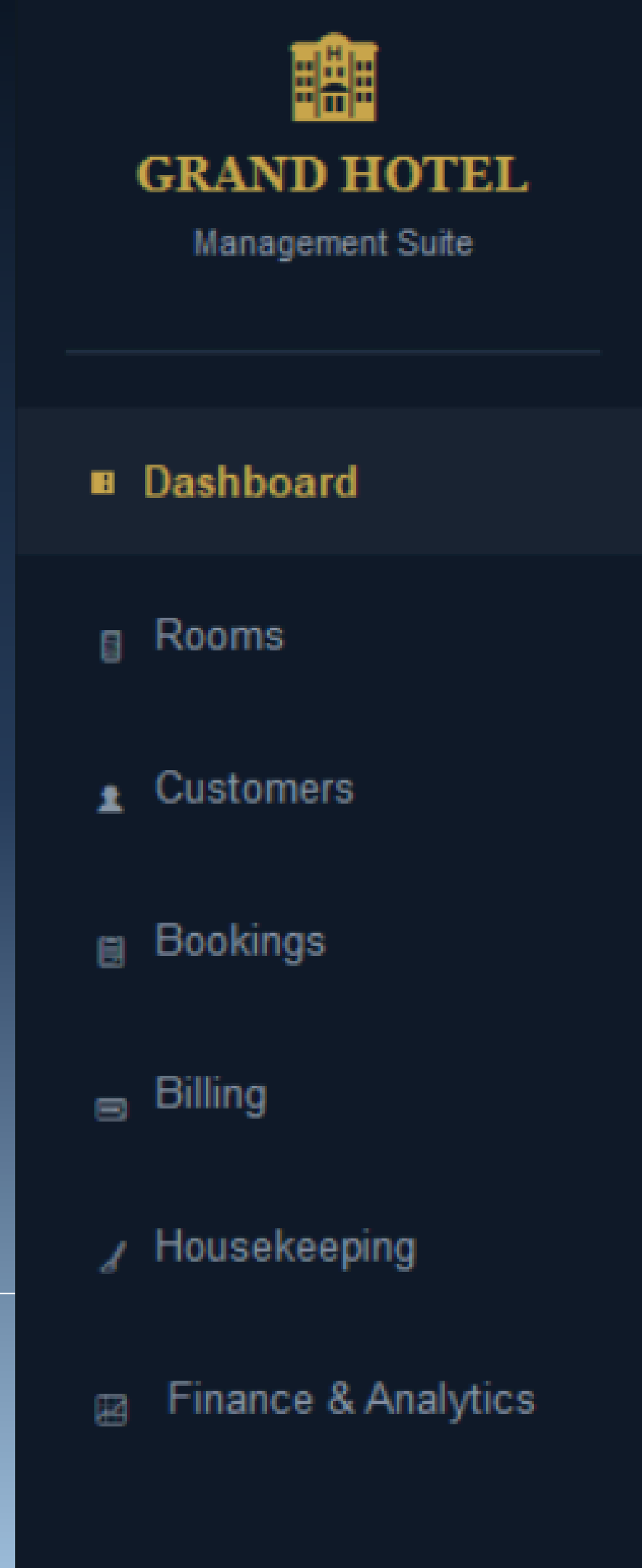


BTECH CSE Batch 66
Python Programming



Objective:

- Automate manual hotel operations
- Improve efficiency & accuracy
- Manage room availability dynamically
- Store and retrieve customer data easily
- Generate bills automatically
- Track revenue and expenses



Technologies Used:

- Python: Core logic and application flow
- Tkinter: GUI components (buttons, forms, tables)
- SQLite: Database management system
- Datetime Module: Date handling (check-in, check-out)
- OOP Concepts: Class-based structure for modular design

```
7  import sqlite3
8  import datetime
9  import tkinter as tk
10 from tkinter import ttk, messagebox, font
11
```



Commands used in creating application:

- `pyinstaller --noconsole --onefile --icon=hotel_icon.ico Project.py`

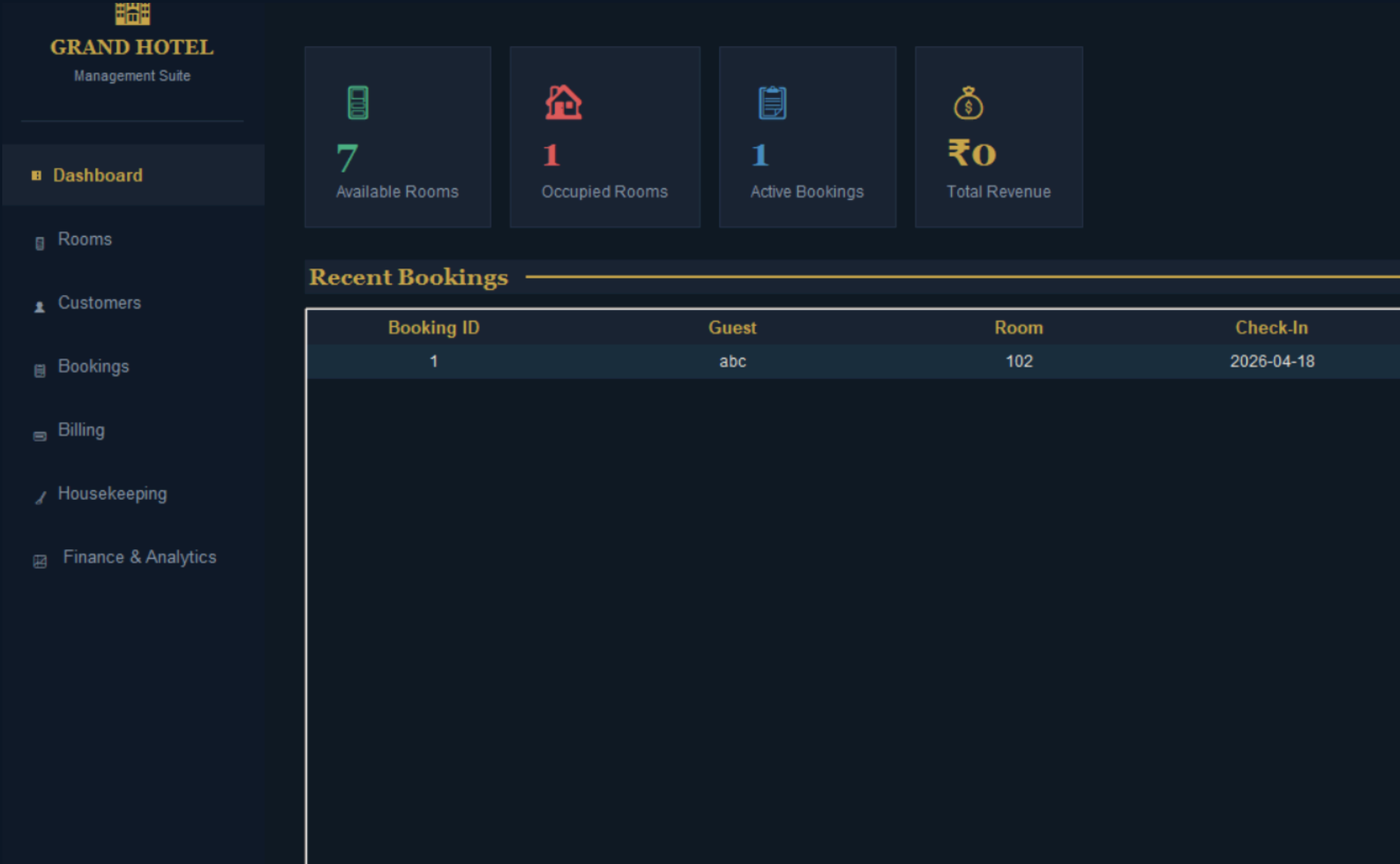
`--noconsole`: This prevents a black command prompt window from popping up when you open your app.

`--onefile`: This packs everything into a single .exe.

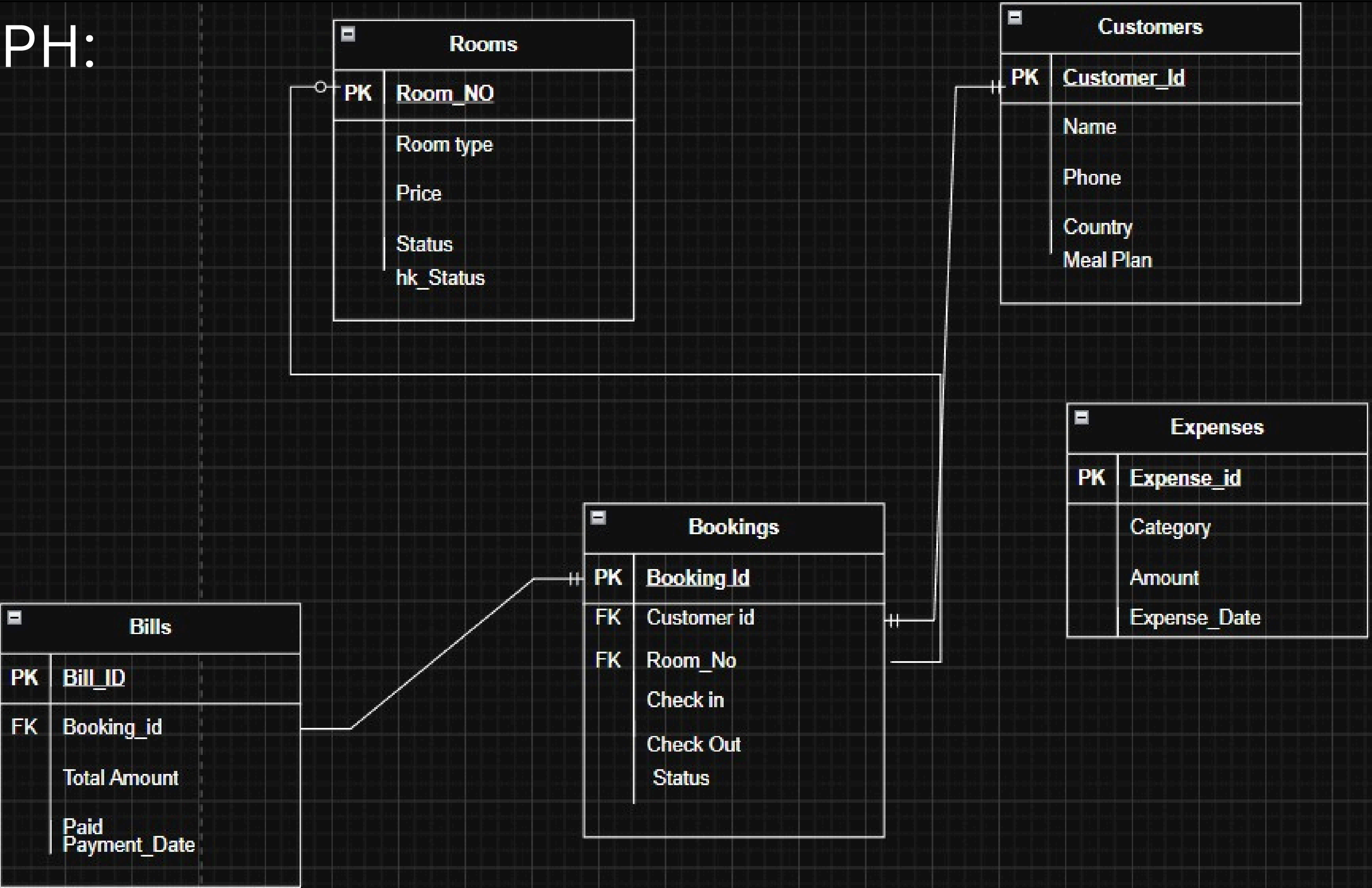
`--icon`: (Optional) If you have an .ico file, this will be the app's logo on the desktop.

SYSTEM MODULES:

- Dashboard (overview & statistics)
- Room Management (add/update rooms)
- Customer Management (store guest details)
- Booking System (reserve rooms)
- Billing System (invoice generation)
- Housekeeping (room status tracking)
- Analytics (revenue & expense reports)



GRAPH:



Challenges Faced:

- Managing relationships between multiple tables (Customers, Rooms, Bookings, Bills)
- Designing a consistent and responsive GUI layout
- Implementing complex billing logic (daily, weekly, monthly pricing)
- Handling real-time updates (room status, checkout, payments)
- Input validation and preventing duplicate/invalid data
- Synchronizing frontend (Tkinter) with backend (SQLite)
- Deployment & Pathing Errors (The "Last Mile")
- SQL Integrity & Logic Errors



SOLUTIONS:

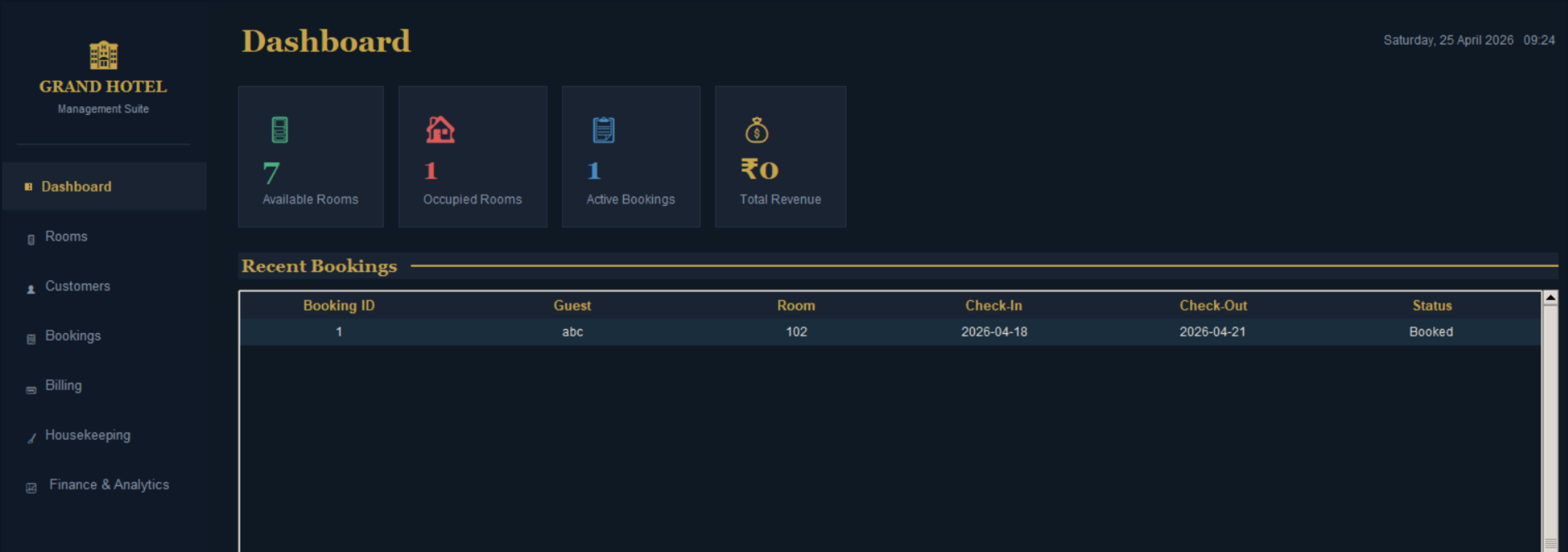
- Used Primary Keys & Foreign Keys to maintain data relationships
- Implemented try-except blocks for error handling
- Created modular functions (e.g., pricing, booking logic)
- Designed reusable UI components for consistency
- Applied step-by-step debugging and testing
- Ensured data validation (phone number, inputs, duplicates)

Room No	Type	Occupancy
101	Single	Available
102	Single	Occupied
103	Double	Available
104	Double	Available
105	Suite	Available
201	Single	Available
202	Double	Available
203	Suite	Available



Output // Conclusion:

- Fully functional hotel management system
- User-friendly graphical interface
- Accurate and automated billing
- Real-time data management



Thank You